

Morning Paper Report - Friday, June 12, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Mechanisms of cognitive change: Training improves the quality but not the quantity of visual working memory representations

Authors unavailable
2023

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory representation, visual search, working memory

Link: https://osf.io/preprints/psyarxiv/xbkcz_v1/

Goal: As of yet, visual working memory (WM) training has failed to yield consistent cognitive benefits on transfer effects to untrained tasks, despite large improvements in trained tasks

Method: N = 64 young healthy adults were randomly assigned to an experimental group or an active control group to complete four training sessions to practise an orientation-reproduction task or visual search, respectively

Results: Therefore, our findings suggest that training gains are not driven by enhanced capacity. Instead, gains in the quality of visual WM representations that are tied to specific stimuli and paradigms may reflect enhanced efficiency in using the existing visual WM capacity.

Conclusion: Therefore, our findings suggest that training gains are not driven by enhanced capacity

2. Perturbing human V1 degrades the fidelity of visual working memory

Mrugank Dake, Clayton E. Curtis
2024

Source: Crossref; matched terms: visual working memory, working memory, fidelity

Link: <https://doi.org/10.1101/2024.06.19.599798>

Goal: Abstract. Decades of macaque research established the importance of prefrontal cortex for working memory

Method: Moreover, concurrently measured electroencephalography confirmed that stimulation disrupted not only memory behavior, but neurophysiological signatures of working memory

Results: These results change the question from whether visual cortex is necessary for working memory to what mechanisms it uses to support memory

Conclusion: These results change the question from whether visual cortex is necessary for working memory to what mechanisms it uses to support memory

3. Comprehensive exploration of visual working memory mechanisms using largescale behavioral experiment

Authors unavailable
2025

Source: OSF/PsyArXiv; matched terms: visual working memory, working memory, neural network, neural networks

Link: https://osf.io/preprints/psyarxiv/6db9p_v3/

Goal: Two decades of research on visual working memory have produced substantial yet fragmented knowledge

Method: Drawing on a large-scale behavioral experiment involving 40 million responses to 10,000 color patterns, a quasi-comprehensive exploration model of visual working memory, termed QCE-VWM, is developed

Results: The model provides an integrative framework for understanding human visual working memory, incorporating a dozen mechanisms—some directly adopted from previous studies, some modified, and others newly identified This work underscores the value of large-scale behavioral experiments in advancing comprehensive models of cognitive mechanisms.

Conclusion: The model provides an integrative framework for understanding human visual working memory, incorporating a dozen mechanisms—some directly adopted from previous studies, some modified, and others newly identified This work underscores the value of large-scale behavioral experiments in advancing comprehensive models of cognitive mechanisms.

4. Comparing memory capacity across stimuli requires maximally dissimilar foils: Using deep convolutional neural networks to understand visual working memory capacity for real-world objects

Timothy F. Brady, Viola S. Störmer
Memory & Cognition, 2024

Source: Crossref; matched terms: visual working memory, working memory, neural network, neural networks

Link: <https://doi.org/10.3758/s13421-023-01485-5>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

5. Modeling visual working memory using recurrent on-center off-surround neural network with distance dependent inhibition

Rakesh Sengupta
Scientific Reports, 2025

Source: Crossref; matched terms: visual working memory, working memory, modeling, neural network

Link: <https://doi.org/10.1038/s41598-025-06919-5>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.